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10. Mai 2012
M/GSH-020-PC/EP
FL/rd

RECOVERY OF HYDROPHOBIC PEPTIDES FROM OILS

New Claims
(marked-up version)

1. A method for separation and recovery of individual cyclic peptides from a plant material, said method comprising the steps of:

extracting an endogenous oil from a plant material selected from a group consisting of seeds, nuts, roots, fruits, fruit pulp, stems, latex, and leaves, wherein the extraction process limits exposure of the endogenous oil to oxygen and the extraction process includes a physical method selected from a group consisting of pressing, grinding, emulsifying, extruding, blending, tempering, exposure to ultrasound, exposure to microwaves, and exposure to infrared radiation;

collecting the extracted oil;

~~separating cyclic peptides from~~ commingling the extracted oil ~~under a condition that limits exposure of the cyclic peptides to oxygen~~ with a suitable absorbent whereby the cyclic peptides attach to the absorbent;

~~recovering individual cyclic peptides from the separated cyclic peptides~~ washing the absorbent at least once with a suitable solvent selected for solubilising at least one plurality of cyclic peptides attached to the absorbent; and

recovering the at least one plurality of cyclic peptides from the solvent.

~~purifying and optionally refining the recovered individual cyclic peptides.~~

2. A method according to claim 1, wherein ~~an~~ selected oil is commingled with the plant material prior to extracting the endogenous oil.

3. A method according to claim 1, wherein the at least one plurality of cyclic peptides ~~are~~ is separated from the extracted oil solubilised using a solvent comprising one of a hydrocarbon ~~selected from a group consisting of propane, butane, hexane and isomers thereof~~, a ketone ~~selected from a group consisting of acetone, 2-butanone and methyl ethyl ketone~~, a lower aliphatic alcohol ~~selected from a group consisting of methanol, ethanol, n-propanol, isopropanol, n-butanol, sec-butanol, t-butanol, amyl alcohol, isoamyl alcohol and active amyl alcohol~~, an ester ~~selected from a group consisting of ethyl acetate and glyceroltriacetate~~, a halogenated solvent ~~selected from a group consisting of chloroform, methylene chloride and carbon tetrachloride~~, an ether ~~selected from a group consisting of dioxan, tetrahydrofuran, dimethyl ether and diethyl ether~~, an aromatic ~~selected from a group consisting of benzene, xylene and toluene~~, and combinations thereof.

4. A method according to claim 3, wherein the hydrocarbon is selected from a group consisting of propane, butane, hexane and isomers thereof, the ketone is selected from a group consisting of acetone, 2-butanone and methyl ethyl ketone, the lower aliphatic alcohol is selected from a group consisting of methanol, ethanol, n-propanol, isopropanol, n-butanol, sec-butanol, t-butanol, amyl-alcohol, isoamyl alcohol and active amyl alcohol, the ester is selected from a group consisting of ethyl acetate and glyceroltriacetate, the halogenated solvent is selected from a group consisting of chloroform, methylene chloride and carbon tetrachloride, the ether is selected from a group consisting of dioxan, tetrahydrofuran, dimethyl ether and diethyl ether, and the aromatic selected from a group consisting of benzene, xylene and toluene.

5. A method according to any one of claims 1 to 4, wherein a series of solvents is sequentially flowed through the adsorbent whereby each solvent is selected for solubilising at least one plurality of cyclic peptides attached to the absorbent.

6. A method according to claim 1, wherein the at least one plurality of cyclic peptides is purified and optionally refined to recover individual cyclic peptides.

47. A method according to claim 6, wherein the individual cyclic peptides are purified cyclic peptides are separated from the extracted oil using a physical method selected from a group consisting of silica-gel column chromatography, high performance liquid chromatography and reversed-phase chromatography.

8. A method according to any one of claims 1 to 7, wherein the suitable absorbent is silica.

9. A method according to claim 8, wherein the silica is washed with a series of increasing polarity solvents, wherein each solvent is selected for solubilisation and elution of specific cyclic peptides.

10. A method according to claim 9, wherein each elution is flowed through chromatography equipment to further separate and purify individual cyclic peptides.

11. A method according to any one of claims 1 to 10, wherein the plant material is flaxseed.

5. A method for separation and collection of individual cyclic peptides from a plant material selected from a group consisting of seeds, nuts, roots, fruits, fruit pulp stems, latex, and leaves, said method comprising the steps of:

- extracting an oil from the plant material;
- recovering a non-polar fraction from the extracted oil;
- separating cyclic peptides from the non-polar fraction;
- recovering of individual cyclic peptides from the separated cyclic peptides; and
- purifying and optionally refining the recovered individual cyclic peptides.

6. — ~~A method according to claim 5, additionally comprising the steps of:~~
~~commingling the extracted oil with an adsorbent whereby at least some cyclic peptides are attached to the adsorbent;~~
~~separating the adsorbent having cyclic peptides attached thereto from the oil;~~
~~flowing therethrough the adsorbent having cyclic peptides attached thereto, at least one solvent selected from a group consisting of water, acetonitrile, ethyl acetate, ethanol, hexane, methanol, dichloromethane, ammonium hydroxide and combinations thereof, whereby at least a plurality of one type of cyclic peptide is recovered into the solvent; and~~
~~separating the plurality of the one type of cyclic peptide from the solvent.~~
7. — ~~A method according to claim 6 wherein a selected plurality of solvents is sequentially flowed through the adsorbent having cyclic peptides attached thereto, whereby each solvent comprising the plurality of solvents recovers at least a plurality of one type of cyclic peptide from the adsorbent.~~
8. — ~~A method according to claim 7, wherein the sequential flow of the plurality of solvents sequentially contacts the adsorbent with a solvent having an increased polarity.~~
9. — ~~A method according to claim 5, wherein the method additionally comprises the steps of:~~
~~commingling the extracted oil with a solution comprising a combination of hydrochloric acid, methanol and water thereby separating the extracted oil into a polar fraction and a non-polar fraction;~~
~~recovering the polar fraction;~~
~~recovering cyclic peptides from the polar fraction with hexane;~~
~~separating individual cyclic peptides from the recovered cyclic peptides; and~~
~~purifying and optionally refining the separated individual cyclic peptides.~~
10. — ~~A cyclolinopeptide having a three dimensional structure according to Figure 18.~~

11. — ~~A cyclolinopeptide recovered by a method according to one of claims 1 and 5, said cyclolinopeptide configured for modulation of a mammalian physiological condition related to apoptosis.~~

12. — ~~A cyclolinopeptide according to claim 11, wherein said physiological condition is selected from a group consisting of tissue damage, neurodegenerative disease, haematologic disease, AIDs, viral infection, inflammatory disease, auto-immune disease and cancer.~~

13. — ~~A cyclolinopeptide according to claim 12, wherein said cancer is selected from a group consisting of colorectal cancer, lung cancer, gliomas, hepatic cancer, neuroblastoma, leukemia, lymphoma and prostate cancer.~~

14. — ~~A cyclolinopeptide according to claim 11, wherein said cyclolinopeptide is selected from a group consisting of cyclolinopeptide A, cyclolinopeptide C, cyclolinopeptide D, cyclolinopeptide E, cyclolinopeptide F and cyclolinopeptide G.~~

15. — ~~A modified flaxseed oil comprising a minimum concentration of at least one plurality of at least one selected cyclolinopeptide, said modified flax oil produced by the steps of:~~

~~separating a flaxseed oil into a polar fraction and a non-polar fraction;~~

~~recovering the non-polar fraction from the extracted oil;~~

~~separating at least one plurality of at least one cyclolinopeptide from the non-polar fraction;~~

~~recovering and quantifying said at least one plurality of the at least one cyclolinopeptide separated from the non-polar fraction;~~

~~commingling a flaxseed oil absent cyclolinopeptides with a selected portion of the recovered at least one plurality of the at least one cyclolinopeptide separated from the non-polar fraction.~~

16. — ~~A modified flaxseed oil comprising a minimum concentration of at least one plurality of at least one selected cyclolinopeptide, said modified flax oil produced by the steps of:~~

~~separating a flaxseed oil into a polar fraction and a non-polar fraction;~~

~~recovering the polar fraction from the extracted oil;
separating at least one plurality of at least one cyclolinopeptide from the polar fraction;~~

~~recovering and quantifying said at least one plurality of the at least one cyclolinopeptide separated from the polar fraction;~~

~~commingling a flaxseed oil absent cyclolinopeptides with a selected portion of the recovered at least one plurality of the at least one cyclolinopeptide separated from the polar fraction.~~

~~17. A modified flaxseed oil comprising a minimum concentration of at least one plurality of at least one selected cyclolinopeptide, said modified flax oil produced by the steps of:~~

~~separating a flaxseed oil into a polar fraction and a non polar fraction;~~

~~recovering the non polar fraction from the extracted oil;~~

~~separating at least one plurality of at least one cyclolinopeptide from the non polar fraction;~~

~~recovering said at least one plurality of the at least one cyclolinopeptide separated from the non polar fraction;~~

~~recovering the polar fraction from the extracted oil;~~

~~separating at least one plurality of at least one cyclolinopeptide from the polar fraction;~~

~~recovering said at least one plurality of the at least one cyclolinopeptide separated from the polar fraction;~~

~~commingling a flaxseed oil absent cyclolinopeptides with at least one of a selected portion of the recovered at least one plurality of the at least one cyclolinopeptide separated from the non polar fraction, and a selected portion of the recovered at least one plurality of the at least one cyclolinopeptide separated from the polar fraction.~~